



OUTLIERS

The Rules

Layer 2 of 8 · Telling one person from another,
which is harder than it sounds.

FOR MAC

EST. MMXXVI · OUTLIERS ACCELERATOR

Who is this?

Every clever feature you add later has to start by asking that question. If it gets the wrong answer, everything after it is wrong too, and nothing reports it.

WHAT THE LAST LAYER LEFT YOU WITH

You have a folder with one file per person in it. It works, and it will keep working right up until the moment 2 files describe the same human being. Nothing in your system currently prevents that, notices it, or could tell you which of the 2 is right.

WHAT THIS LAYER BUILDS

This layer gives your CRM a way to know that one person is one person, however their name happens to be written down that day. It also writes down the shape a record has to follow, in a form software can check, so a broken record gets refused as it is written rather than found months later.

ONE PERSON, 4 WAYS OF WRITING THEM DOWN



Match on the name instead, and 2 different people who share one get merged into a single record.

4 WAYS IN. ONE PERSON OUT.

How it works

2 ideas, and the first one is the whole layer.

1. ONE IDENTIFIER DECIDES WHO SOMEONE IS

A name is not an identifier. Names repeat, people change them, and the same person gets written down 4 different ways by 4 different sources. What you need is one detail per person that is unique to them and does not change: usually a profile link, sometimes an email address.

Your system now picks that identifier first and treats everything else as decoration. Every other spelling of their name is kept in a list beside it, so when any of those spellings turns up later, the system knows where it belongs.

Why it never guesses. If a name could be 2 different people, the system refuses to answer rather than picking one. That is deliberate. Consider the alternative: a wrong merge welds 2 people's histories together, silently and permanently, and you will not find out until you say the wrong words to someone.

2. THE SHAPE OF A RECORD, WRITTEN DOWN WHERE SOFTWARE CAN CHECK IT

A rule that nothing checks will be broken without anyone noticing. Your system now keeps a short description of what a record must contain, and anything writing to your CRM has to satisfy it first. A record that is missing something gets refused at the moment it is written.

The same file also keeps the list of words your system is allowed to use for each idea, so one idea never ends up with 5 names for it. It sounds fussy. It is what makes the difference between a CRM you can ask questions of and a folder of notes.

AND 3 RULES IT NEVER BREAKS

- **Never delete a person.** People come back. Records get set aside, not destroyed.
- **Never store anything sensitive** about someone's health, beliefs or politics.
- **Never let an automation speak as you.** That one becomes machinery in Layer 6.

Before you start on a Mac

Everything in this guide is typed into **Terminal**, the Mac's window for typing commands instead of clicking. Press Command and Space together, type Terminal, and press Return.

This layer needs Python and git, which you set up for Layer 1 (its page "Before you start on a Mac"), and the layers before this one. It installs no add-ons, so it works with either Python a Mac may have.

HOW TO TYPE THE LINES

Where this guide shows several lines in a black box, type them one at a time: type the first, press Return, wait for it to finish and give you a fresh line, then type the next. The installer asks you questions. Each offers an answer in [square brackets]; press Return to take it, or type your own and press Return.

WHERE YOUR CRM IS

Layer 1 made your CRM at ~/CRM: a folder called CRM in your home folder (the folder with your name and the house picture in Finder's sidebar). If you chose another folder in Layer 1, put your own folder wherever a line in this guide says ~/CRM. The installer finds your CRM by itself either way.

Apple Silicon or Intel (the 2 kinds of chip a Mac can have; the Apple menu, then About This Mac, shows yours): the steps are the same on both, and both were tested.

Tried only on test Macs. Every step in this guide and in the instructions file of the layer's download was tried on test Macs (Macs GitHub rents out by the minute to run a script, not a person's own Mac, and not yours), never on a real Mac, except 2: the box asking to install the developer tools, which cannot appear on a test Mac; **python3 identity.py who**, because you type it with the nickname you add to the record yourself. The last pages list every line and what happened to it.

Build it

3 lines, typed into Terminal. The last one asks you a few questions so what you get is yours, then does the work.

```
git clone https://github.com/OUTLIERS-ai/outliers-crm-02-rules-mac
cd outliers-crm-02-rules-mac
python3 install.py
```

This layer needs Layer 1 already installed. It finds your CRM by itself and will tell you politely if it cannot.

WHAT IT ASKS YOU, AND WHY

QUESTION	WHAT IT CHANGES
Which detail will you always have for someone?	A profile link, an email address, or a phone number. Whichever you always end up with. This becomes how your system tells one person from another.
Do you keep personal contacts in here too?	Friends and family often end up in the same system. If you say yes, they get marked as personal, and nothing commercial will ever touch them.

WHAT YOU END UP WITH

- A written description of what a record must contain, in a file you can read and change.
- A tool that answers **who is this?** for any name, link or address you give it.
- Every record now carries the list of other names that same person is known by.

Now use it

Ask your system who somebody is, 2 different ways, and watch one record come back.

Give your system the same person twice, written 2 different ways. Use their full name one time and their profile link the other. Use a person of your own whose record has a profile link. The lines below use Rowan Ashdown, the made-up person Layer 1 put in your CRM, with a made-up link; your Rowan has no link, on purpose, so put your own person's name and link in place of these.

The tools this layer installed live in a folder called **_engine** inside your CRM. In Terminal, go there first, then type the next 2 lines (if your CRM is not at ~/CRM, put your own folder in the cd line):

```
cd ~/CRM/_engine
python3 identity.py who "Rowan Ashdown"
python3 identity.py who "https://www.linkedin.com/in/rowan-ashdown-000"
```

Both print the same record. Now open that person's file in Obsidian. Near the top is a line beginning **aliases:**. Add their nickname on a new line beneath it, save, and run **python3 identity.py who** again with the nickname. It finds them.

Every layer above this one depends on doing that reliably, thousands of times, without a human checking.

IF YOU ALREADY HAD SOMETHING LIKE THIS

A test rather than an instruction. Take 2 records in your existing system that you suspect are the same person, and work out which one is correct. If you cannot do that quickly and confidently, this layer is doing real work for you.

WHAT THIS LAYER LEAVES UNSOLVED

Your records now have a shape and an identity, but nothing fills them in. Worse, the obvious details to write down about a person go out of date on their own: someone's job title is correct right up until they change roles, and nothing in the record knows that happened.

NEXT: LAYER 3 • WHAT YOU STORE

Recording what happened instead of what is true today, so your CRM stops going stale the moment you look away from it.

WORDS USED IN THIS GUIDE

Identifier	One detail that belongs to a person and does not change. Usually a profile link.
Alias	Any other name or address the same person is known by.
Schema	A written description of what a record must contain, in a form software can check.
Record	One person's file in your CRM.

What was run on a Mac

A **test Mac** is one of the Macs GitHub rents out by the minute to run a script.

Each line below is one this guide or the instructions file of the layer's download (README.md) gives you to type.

Before the 2 lines that look a person up, Rowan Ashdown, the made-up person Layer 1 puts in every new CRM, was given the made-up profile link those lines use, so both had a record to find.

The run was on 2026-09-26. A script (a list of commands each test Mac carries out by itself) downloaded a fresh copy of the download from GitHub on each of 7 test Macs and ran each line below exactly as printed, in this order. After a `cd` line (`cd` moves Terminal into a folder), the lines that follow ran in that folder, as they do for you. The script ran them; no person typed them.

7 test runs: 3 on Macs set up from scratch (macOS 15.7.9 on Apple Silicon, macOS 15.7.9 on Intel, macOS 26.6.2 on Apple Silicon), and 4 on 2 Macs that also have Homebrew, an add-on installer many Mac owners use, each run once with Homebrew installed before Python and once after (macOS 15.7.9 on Apple Silicon and macOS 15.7.9 on Intel). Before the first line, the script put these on every Mac: Python from python.org; Layer 1 of the CRM, installed from its Mac download. GitHub's Macs already have Apple's developer tools, which include git.

EACH LINE, THEN WHAT HAPPENED TO IT

```
git clone https://github.com/OUTLIERS-ai/outliers-crm-02-rules-mac
```

worked on all 7 test Macs

```
cd outliers-crm-02-rules-mac
```

worked on all 7 test Macs

```
python3 install.py
```

worked on all 7 test Macs

```
cd ~/CRM/_engine
```

worked on all 7 test Macs

```
python3 identity.py who "Rowan Ashdown"
```

worked on all 7 test Macs

```
python3 identity.py who "https://www.linkedin.com/in/rowan-ashdown-000"
```

worked on all 7 test Macs

```
python3 identity.py who
```

not run in this test: you type it with the nickname you add to the record yourself

```
cd ~/CRM
```

worked on all 7 test Macs

EACH LINE, CONTINUED

```
python3 _engine/identity.py stats
```

worked on all 7 test Macs

```
python3 _engine/identity.py who "a name"
```

worked on all 7 test Macs

```
python3 _engine/identity.py duplicates
```

worked on all 7 test Macs

```
python3 _engine/identity.py collisions
```

worked on all 7 test Macs

```
python3 _engine/schema.py contract
```

worked on all 7 test Macs

```
python3 _engine/schema.py sweep
```

worked on all 7 test Macs

```
cd ~/outliers-crm-02-rules-mac
```

worked on all 7 test Macs

```
python3 tests/test_identity_resolution.py
```

worked on all 7 test Macs

```
python3 tests/test_schema_contract.py
```

worked on all 7 test Macs

The script's record of the run is public, for anyone who wants to check it; you do not need to open it:

<https://github.com/OUTLIERS-ai/outliers-mac-test/actions/runs/36210549122>

NOT RUN ON THE TEST MACS

The box asking to install the developer tools cannot appear on a test Mac, because GitHub's Macs already have them.

Opening your CRM in Obsidian. The test Macs have no Obsidian; the script read the folders and files the installer made instead.